

Ruiqi Rae Yu

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Education

Ph.D. in Operations Research & Financial Engineering 2021 – 2026
Princeton University

- Topics: Mathematical Statistics, Causal Inference
- Advisor: Matias Cattaneo, Department of Operations Research & Financial Engineering

M.A. in Operations Research & Financial Engineering 2021 – 2023
Princeton University

Honors Bachelor of Science in Mathematics 2017 – 2021
University of Toronto

Research & publications

Publications

1. Strong approximations for empirical processes indexed by Lipschitz functions, with M. D. Cattaneo. *The Annals of Statistics*, 53(3), 1203–1229, 2025.

Working papers

1. Estimation and inference in boundary discontinuity designs, with M. D. Cattaneo and R. Titiunik. *arXiv preprint*, 2025 [arXiv:2505.05670](https://arxiv.org/abs/2505.05670).
2. Causal treatment effect aggregation in boundary discontinuity designs, with M. D. Cattaneo and R. Titiunik. [URL for current version](#), 2025.
3. Robust inference for the direct average treatment effect with treatment assignment interference, with M. D. Cattaneo, Y. He. *arXiv preprint*, 2025. [arXiv:2502.13238](https://arxiv.org/abs/2502.13238). Submitted to *Biometrika*.
4. The Honest Truth About Causal Trees: Accuracy Limits for Heterogeneous Treatment Effect Estimation, with M. D. Cattaneo and J. M. Klusowski. *arXiv preprint*, 2025. [arXiv:2509.11381](https://arxiv.org/abs/2509.11381). Submitted to *PNAS*.

Software articles

1. rd2d: causal inference in boundary discontinuity designs, with M. D. Cattaneo and R. Titiunik. *arXiv preprint*, 2025. [arXiv:2505.07989](https://arxiv.org/abs/2505.07989).
[rd2d \(CRAN\)](#): Companion R package.

Teaching

Princeton University Sep 2022 – Dec 2025

- ORF 524: Statistical Theory and Methods, Fall 2025
- ORF 245: Fundamentals of Statistics, Fall 2022, Spring 2023, Fall 2024

University of Toronto Sep 2019 – Dec 2020

- MAT 244: Ordinary Differential Equations, Fall 2020
- MAT 137 Calculus, Spring 2019
- MAT 223: Linear Algebra, Fall 2018

Awards & funding

- DDSS Graduate Fellowship in Social Data Science, Princeton University 2024, 2025
- Coxeter Scholarship in Mathematics, University of Toronto 2021, 2019
- University of Toronto Scholar – Rogers, University of Toronto 2019
- Adel S Sedra Undergraduate Scholarship in Mathematics, University of Toronto 2018

Services

Peer review

Econometric Theory; Economic Letters; Journal of the American Statistical Association; Journal of Causal Inference; Journal of Econometrics; Operations Research; Statistical Science

Presentations & conferences

- Joint Statistical Meeting, Invited Paper Session, August 2025.
- Data-Driven Social Science Connect, Princeton University, November 2024, April 2024.